

SECURE QR CODE MESSAGE TRANSMISSION SYSTEM

Python Source Code

Appendix: Source code used for AES-256 encryption, TCP communication, QR generation, OTP verification, and decryption.

sender.py

```
import socket
import base64
from cryptography.hazmat.primitives.ciphers.aead import AESGCM

key = b"0123456789abcdef0123456789abcdef"

message = input("Enter the message to send: ")

aes = AESGCM(key)
nonce = __import__("os").urandom(12)
ciphertext = aes.encrypt(nonce, message.encode(), None)

encrypted_data = base64.b64encode(nonce + ciphertext).decode()

print("\nEncrypted Base64 data:")
print(encrypted_data)

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("127.0.0.1", 5000))
client.sendall(encrypted_data.encode())
client.close()
```

receiver.py

```
import socket
import base64
from cryptography.hazmat.primitives.ciphers.aead import AESGCM

key = b"0123456789abcdef0123456789abcdef"

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("127.0.0.1", 5000))
server.listen(1)

print("Receiver is waiting for encrypted data...")
print("Listening on TCP port 5000...")

conn, addr = server.accept()
data = conn.recv(4096)
conn.close()
server.close()

encrypted_data = data.decode()
print("\nEncrypted Base64 data received:")
print(encrypted_data)

raw = base64.b64decode(encrypted_data)
nonce = raw[:12]
ciphertext = raw[12:]

aes = AESGCM(key)
decrypted_message = aes.decrypt(nonce, ciphertext, None)

print("\nDecrypted message:")
print(decrypted_message.decode())
```

qr_generator.py

```
import qrcode
import secrets

encrypted_data = input("Paste the encrypted Base64 data: ")

otp = str(secrets.randbelow(900000) + 100000)

qr = qrcode.make(encrypted_data)
qr.save("encrypted_message_qr.png")

print("\n===== SECURE QR SYSTEM =====")
print("QR code generated successfully!")
print("Saved as: encrypted_message_qr.png")
print("OTP for authorized receiver:", otp)
print("=====")

with open("otp.txt", "w") as file:
    file.write(otp)

print("\nSend the OTP separately to the receiver.")
```

qr_decoder.py

```
import base64
from cryptography.hazmat.primitives.ciphers.aead import AESGCM

key = b"0123456789abcdef0123456789abcdef"

with open("otp.txt", "r") as file:
    correct_otp = file.read().strip()

print("\n===== SECURE QR MESSAGE =====")
print("Enter the OTP received separately.")
print("=====")

encrypted_data = input("\nPaste the encrypted data from the QR code: ")
entered_otp = input("Enter OTP: ")

if entered_otp != correct_otp:
    print("\nOTP verification failed!")
    print("Message cannot be decrypted.")
    exit()

print("\nOTP verified successfully!")

data = base64.b64decode(encrypted_data)
nonce = data[:12]
ciphertext = data[12:]

aes = AESGCM(key)
decrypted_message = aes.decrypt(nonce, ciphertext, None)

print("\nDecrypted message:")
print(decrypted_message.decode())
```